Module 3: Introduction to the simulation study

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Description

How does a simulation study work?

Prerequisites

- Module 'Introduction to simulation'
- Module 'Decide whether to simulate'

Learning Objectives

You can:

- name the phases of a simulation study.
- record the course of a simulation study with the various contexts.
- identify the problems and risks involved in carrying out a simulation study.

Time Requirement

- 30 - 40 minutes

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Overview

What you can expect in this learning unit

In this learning unit, you will learn step by step how to systematically carry out a simulation study. The knowledge from the previous modules is taken up and expanded here. We will not only show you which phases and activities a simulation study comprises, but also how to apply them in practice and which potential pitfalls to avoid.

Your journey through the simulation study:

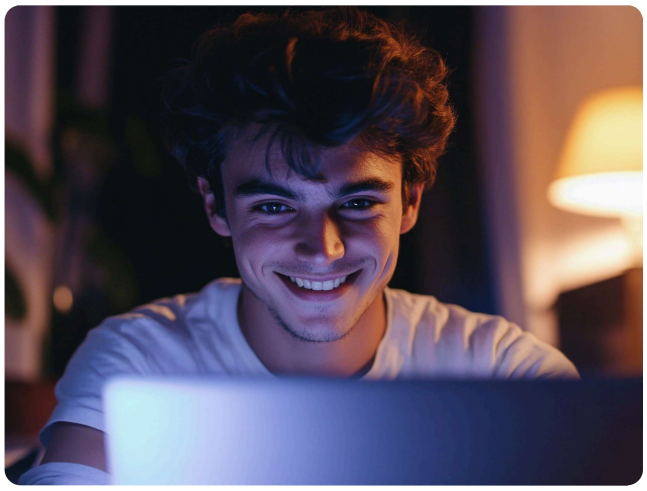
- **Understand:** Get to know the individual phases and understand how they build on each other.
- **Visualize:** Draw the process and internalize the connections.
- **Recognize:** Identify problems that can occur during implementation and learn how to solve them.

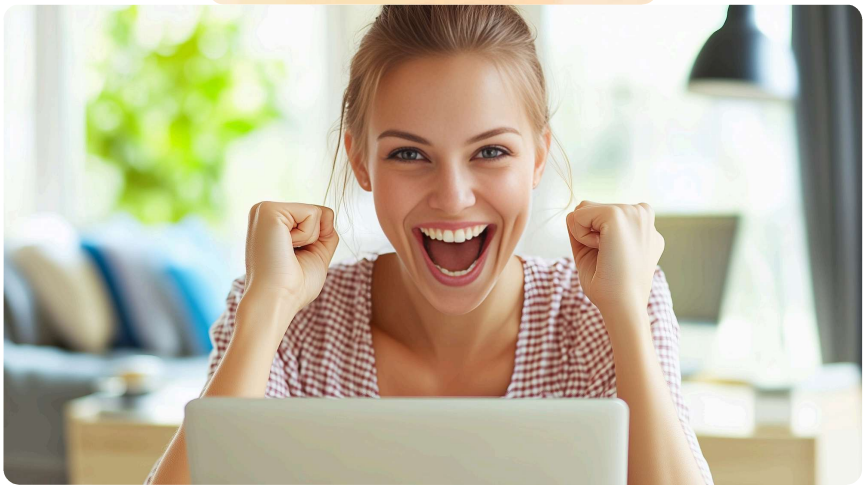
Why is this learning unit worthwhile?

- With the knowledge gained from this unit, you will be able to analyze complex systems and make well-founded decisions - skills that are in demand in many areas!

How are you supported?

- You learn with a combination of illustrations, reflection questions, clear videos and a structured guide.





Ready to take your first step into the world of simulation studies? Let's go - with joy and a clear goal in mind!

Let's get started!

Introduction

The structured path to a successful simulation study

A simulation study is much more than simply creating a model - it requires a clear structure and a methodical approach in order to achieve reliable results. The aim is to realistically depict complex systems and gain valuable insights for well-founded decisions. This structured process ensures that no important factors are overlooked and that the results remain reproducible.

What is a simulation study?

A simulation study is a structured process that involves the application of simulation methods to analyze and evaluate complex systems or processes.

These studies are widely used in various fields, including engineering, business administration, healthcare, production, and logistics, where efficiency and effectiveness are of utmost importance.

The goal is to create realistic models to understand and optimize the behavior of systems under different conditions.

Systematic Approach for a Simulation Study

Once it has been decided that simulation will be applied to the problem, a simulation study is conducted.

Carrying out a simulation study requires a systematic approach. This means following a clearly structured and methodical process to solve a problem. In the context of a simulation study, this includes:

- **Step-by-step planning:** A predefined sequence of steps is followed to ensure that all necessary aspects are taken into account.

- **Documentation:** Each step is recorded to ensure transparency and to be able to refer back to decisions made and data collected later on.
- **Structured analysis:** The system and the relevant data are thoroughly analyzed before the modeling process begins.
- **Validation and adjustment:** The model is systematically checked and tested to ensure that it produces realistic results. Adjustments are made if necessary.
- **Traceability:** Each step of the approach should be logical and traceable, so that others can understand the process and reproduce the results.

This methodological approach minimizes errors, improves the quality of results, and ensures that all relevant factors are taken into account in the simulation study.

A simulation study follows systematic phases. **Figure 1** shows the various phases of a simulation study, from problem definition to modeling and analysis of results. It serves as a guideline to structure and conduct the process of a simulation study systematically and efficiently.

Figure 1: Phases of the simulation study with steps in detail

The next chapter provides a more detailed explanation of the individual phases of the simulation study, as shown in **Figure 1**. This will give you a deeper insight into the structured process and the key steps of a successful simulation study.

The Phases of a Simulation Study

From Problem Analysis to Documentation

A successful simulation study typically begins with the problem analysis and the definition of the objective, followed by data preparation and system modeling, the execution of simulations, and the analysis of results.

Figure 2 illustrates the individual phases that are followed when conducting a simulation study.

Figure 2: Phases of a simulation study with details

Figure 2 is very comprehensive and serves as a helpful guide throughout a simulation study.

Take a look at Figure 2 and try to understand it.

If you don't fully understand everything yet, Video 1 should help you further. However, learning from concrete examples is often more effective than learning theoretical concepts. Let's now move from theory to practice! In **Video 1**, the simulation phases are explained using a small example.

*Watch **Video 1** carefully. Pay special attention to the individual phases of a simulation study and how they are practically implemented using the example of a food laboratory.*

Observe how problems are analyzed, objectives are defined, and models are developed and validated.

*After watching **Video 1**, answer the following questions to check your knowledge and understanding.*

The video presentation is in English.

Video 1: Introduction to the Simulation Study using an Example by Dr. Tatiana Starostina (12:27)

Follow-up questions to video 1

What results can be derived from simulation experiments?

An overview of the exact costs of the system

The final decision on the system design

A report on the technical limitations

✔ Statistics such as key figures, boxplots and diagrams

Correct answer: The video explains that the results of simulation experiments are analysed using statistics and graphical representations.

1/1

Finish

Use these questions to reflect on your knowledge of the simulation phases and their practical implementation. Think about how you could apply the presented methods in other contexts.

You seem to have understood the phases of a simulation study so far. Now you are ready to learn even more about the approach.

Let’s move on to the next chapter!

The Approach to Conducting a Simulation Study

A Structured Approach to Conducting Simulation Studies

The detailed approach for conducting a simulation study is shown in **Figure 3** and compared with **Figure 2** from the previous chapter.

A key aspect is that the individual phases and activities are not viewed in isolation, but are closely interconnected. Feedback loops play a crucial role in this context, as they highlight the iterative nature of the process.

These feedback loops make it possible to detect errors at an early stage and gradually improve the model.

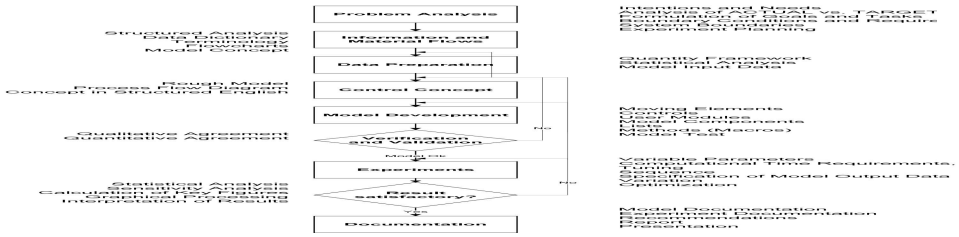


Figure 3: Procedure for simulation (Burkhard, 2003)

By repeatedly carrying out this process, the representation of the real system is continuously refined and adjusted to reality. This iterative process is essential for increasing the quality and significance of the simulation.

*Take a look at **Figure 3** and then answer the reflection questions below.*

Reflection questions on the course of a simulation study

After the model was developed and validated, it turned out that some results of the simulation do not match reality.

In this case, which activities can help to bring the results closer to reality?

Select other activities that can help to bring the results closer to reality.

- Increase the complexity of the model without objectives
- ✔ Check input data
- ✔ consider what influence the level of detail has on the results
- Set all model parameters to default values
- ✔ Increase the level of detail
- Accept simulation results without further comparison with reality

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Finish

After learning about the phases of a simulation study and the corresponding approach, you should be able to answer a few questions about them.

In the next and final chapter, you can test your knowledge and complete this digital learning unit.

Completion of the Learning Unit

Answer the following questions to complete the module.

You are now familiar with the phases and activities of a simulation study. You should be able to answer the following questions.

Final exam question

Specific activities can be assigned to each phase of a simulation study.

Frage zu den Phasen einer Simulationsstudie

If you answer the question correctly, you will receive a badge of distinction. This badge will be visible at the top of the course. With this badge, you can unlock and start additional self-study learning units online.

Enjoy!